

AIAC Assignment 3.5

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Question 1: Zero-Shot Prompting (Leap Year Check)

Write a zero-shot prompt to generate a Python function that checks whether a given year is a leap year.

Task:

- Record the AI-generated code.
- Test with years like 1900, 2000, 2024.
- Identify logical flaws or missing conditions.

Prompt: write a python code to check if the given year is leap year or not

Code:

```
def is_leap_year(year):  
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):  
        return True  
    else:  
        return False  
  
# Example usage  
year = int(input("Enter a year: "))  
if is_leap_year(year):  
    print(f"{year} is a leap year.")  
else:  
    print(f"{year} is not a leap year.")
```

```
C: > Users > ASHWITHA > Downloads > leapyr.py > ...
1  # write a python code to check if the given year is leap year or not
2  def is_leap_year(year):
3      if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
4          return True
5      else:
6          return False
7  # Example usage
8  year = int(input("Enter a year: "))
9  if is_leap_year(year):
10     print(f"{year} is a leap year.")
11 else:
12     print(f"{year} is not a leap year.")
13
```

Output:

```
Enter a number: 6
6 is an even number.
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASHWITHA/Downloads/even.py
Enter a number: a
Traceback (most recent call last):
  File "c:\Users\ASHWITHA\Downloads\even.py", line 7, in <module>
    number = int(input("Enter a number: "))
ValueError: invalid literal for int() with base 10: 'a'
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASHWITHA/Downloads/leapyr.py
Enter a year: 2002
2002 is not a leap year.
PS C:\Users\ASHWITHA>
```

Question 2: One-Shot Prompting (GCD of Two Numbers)

Write a one-shot prompt with one example to generate a Python function that finds the Greatest Common Divisor (GCD) of two numbers.

Example:

Input: 12, 18 → Output: 6

Task:

- Compare with a zero-shot solution.
- Analyze algorithm efficiency

Prompt: """

Input: 12, 18 → Output: 6

"""

Code:

```
def gcd(a, b):
    while b:
        a, b = b, a % b
    return a

# Example usage
num1 = int(input("Enter first number: "))

num2 = int(input("Enter second number: "))

result = gcd(num1, num2)

print(f"The GCD of {num1} and {num2} is {result}.")
```

```
C: > Users > ASHWITHA > Downloads > leapyr.py > ...
1  """
2  Input: 12, 18 → Output: 6
3  """
4  def gcd(a, b):
5      while b:
6          a, b = b, a % b
7      return a
8  # Example usage
9  num1 = int(input("Enter first number: "))
10
11  num2 = int(input("Enter second number: "))
12  result = gcd(num1, num2)
13  print(f"The GCD of {num1} and {num2} is {result}.")
14
```

Output:

```
Traceback (most recent call last):
  File "c:/Users/ASHWITHA/Downloads/leapyr.py", line 7, in <module>
    number = int(input("Enter a number: "))
ValueError: invalid literal for int() with base 10: 'a'
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASHWITHA/Downloads/leapyr.py
Enter a year: 2002
2002 is not a leap year.
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASHWITHA/Downloads/leapyr.py
Enter first number: 17
Enter second number: 44
The GCD of 17 and 44 is 1.
PS C:\Users\ASHWITHA>
```

Question 3: Few-Shot Prompting (LCM Calculation)

Write a few-shot prompt with multiple examples to generate a Python function that computes the Least Common Multiple (LCM).

Examples:

- Input: 4, 6 → Output: 12
- Input: 5, 10 → Output: 10
- Input: 7, 3 → Output: 21

Task:

- Examine how examples guide formula selection.
- Test edge cases.

Prompt: """calculate the Least Common Multiple (LCM) of two given numbers.

Input: 4, 6 → Output: 12

Input: 5, 10 → Output: 10

Input: 7, 3 → Output: 21

"""

Code:

```
def lcm(a, b):
```

```
    def lcm(x, y):
```

```
        while y:
```

```
            x, y = y, x % y
```

```
        return x
```

```
    return abs(a * b) // lcm(a, b)
```

```
# Example usage
```

```
num1 = int(input("Enter first number: "))
```

```
num2 = int(input("Enter second number: "))
```

```
print(f"The LCM of {num1} and {num2} is {lcm(num1, num2)}")
```

C:\Users\ASHWITHA> Downloads > leapyr.py > ...

```
1  """calculate the Least Common Multiple (LCM) of two given numbers.
2  Input: 4, 6 → Output: 12
3  Input: 5, 10 → Output: 10
4  Input: 7, 3 → Output: 21
5  """
6  def lcm(a, b):
7      def lcm(x, y):
8          while y:
9              x, y = y, x % y
10         return x
11     return abs(a * b) // lcm(a, b)
12 # Example usage
13 num1 = int(input("Enter first number: "))
14 num2 = int(input("Enter second number: "))
15 print(f"The LCM of {num1} and {num2} is {lcm(num1, num2)}")
16 |
```

Output:

```
Enter first number: 17
Enter second number: 44
The GCD of 17 and 44 is 1.
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe
Enter first number: 17
Enter second number: 44
The LCM of 17 and 44 is 748
PS C:\Users\ASHWITHA> |
```

Question 4: Zero-Shot Prompting (Binary to Decimal Conversion)

Write a zero-shot prompt to generate a Python function that converts a binary number to decimal.

Task:

- Test with valid and invalid binary inputs.
- Identify missing validation logic

Prompt: write a python code to convert binary number to decimal number

Code:

```
def binary_to_decimal(binary_str):
    decimal_number = 0
    binary_str = binary_str[::-1] # Reverse the string to process from least significant bit
    for index, digit in enumerate(binary_str):
        if digit == '1':
```

```

        decimal_number += 2 ** index

    return decimal_number

# Example usage

binary_input = input("Enter a binary number: ")

decimal_output = binary_to_decimal(binary_input)

print(f"The decimal equivalent of binary {binary_input} is {decimal_output}.")

```

```

C: > Users > ASHWITHA > Downloads > leapyr.py > ...
1  #write a python code to convert binary number to decimal number
2  def binary_to_decimal(binary_str):
3      decimal_number = 0
4      binary_str = binary_str[::-1] # Reverse the string to process from least significant bit
5      for index, digit in enumerate(binary_str):
6          if digit == '1':
7              decimal_number += 2 ** index
8      return decimal_number
9  # Example usage
10 binary_input = input("Enter a binary number: ")
11 decimal_output = binary_to_decimal(binary_input)
12 print(f"The decimal equivalent of binary {binary_input} is {decimal_output}.")
13

```

Output:

```

The LCM of 17 and 44 is 748
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASHWITHA/Downlo
Enter a binary number: 010001
The decimal equivalent of binary 010001 is 17.
PS C:\Users\ASHWITHA>

```

Question 5: One-Shot Prompting (Decimal to Binary Conversion)

Write a one-shot prompt with an example to generate a Python function that converts a decimal number to binary.

Example:

Input: 10 → Output: 1010

Task:

- Compare clarity with zero-shot output.
- Analyze handling of zero and negative numbers.

Prompt:

"""

Input: 10 → Output: 1010

"""

Code:

```
def int_to_binary(n):  
    if n == 0:  
        return "0"  
    binary_str = ""  
    while n > 0:  
        binary_str = str(n % 2) + binary_str  
        n = n // 2  
    return binary_str  
  
# Example usage  
number = int(input("Enter a number: "))  
binary_representation = int_to_binary(number)  
print(f"The binary representation of {number} is {binary_representation}.")
```

C: > Users > ASHWITHA > Downloads > leapyr.py > ...

```
1  """  
2  Input: 10 → Output: 1010  
3  """  
4  def int_to_binary(n):  
5      if n == 0:  
6          return "0"  
7      binary_str = ""  
8      while n > 0:  
9          binary_str = str(n % 2) + binary_str  
10         n = n // 2  
11     return binary_str  
12 # Example usage  
13 number = int(input("Enter a number: "))  
14 binary_representation = int_to_binary(number)  
15 print(f"The binary representation of {number} is {binary_representation}.")  
16 |
```

Output:

SyntaxError: invalid syntax

```
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASHWITHA  
Enter a number: 123456  
The binary representation of 123456 is 11110001001000000.  
PS C:\Users\ASHWITHA> |
```

Question 6: Few-Shot Prompting (Harshad Number Check)

Write a few-shot prompt to generate a Python function that checks whether a number is a Harshad (Niven) number.

Examples:

- Input: 18 → Output: Harshad Number
- Input: 21 → Output: Harshad Number
- Input: 19 → Output: Not a Harshad Number

Task:

- Test boundary conditions.
- Evaluate robustness

Prompt:

```
""""  
  
Input: 18 → Output: Harshad Number  
Input: 21 → Output: Harshad Number  
Input: 19 → Output: Not a Harshad Number  
  
""""
```

Code:

```
def is_harshad_number(n):  
    digit_sum = sum(int(digit) for digit in str(n))  
    return n % digit_sum == 0  
  
# Example usage  
number = int(input("Enter a number: "))  
if is_harshad_number(number):  
    print(f"{number} is a Harshad Number.")  
else:  
    print(f"{number} is Not a Harshad Number.")
```



```
C: > Users > ASHWITHA > Downloads > leapyr.py > ...
1  """
2  Input: 18 → Output: Harshad Number
3  Input: 21 → Output: Harshad Number
4  Input: 19 → Output: Not a Harshad Number
5
6  """
7
8  def is_harshad_number(n):
9      digit_sum = sum(int(digit) for digit in str(n))
10     return n % digit_sum == 0
11
12     # Example usage
13     number = int(input("Enter a number: "))
14     if is_harshad_number(number):
15         print(f"{number} is a Harshad Number.")
16     else:
17         print(f"{number} is Not a Harshad Number.")
```

Output:

```
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASHWITHA/
Enter a number: 1234
1234 is Not a Harshad Number.
PS C:\Users\ASHWITHA> & C:/Users/ASHWITHA/AppData/Local/Programs/Python/Python314/python.exe c:/Users/ASHWITHA/
Enter a number: 4567
4567 is Not a Harshad Number.
PS C:\Users\ASHWITHA> 
```